

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0870

Commonwealth Chesapeake 345 kV
September 18, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0870. The Interconnection Customer proposes a 937.6 MW natural gas combined cycle facility interconnecting at the Commonwealth Chesapeake 345 kV substation in the Virginia area. The request was received on October 04, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 261 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$88,577,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Vienna Operations 138/1kV TX	Easton 2 to Vienna Operations 345.0 kV	282.8	452	50.0
Southtown to Vienna Operations 345.0 kV	St Bonifacius to W -Station 138.0 kV Line	275.2	1,020	92.7
Valleyview to Prunedale Substation 138.0 kV	Southtown to Commonwealth Chesapeake 345.0 kV	219.2	235	20.6
Henning to Kiwi 138.0 kV	Vienna Operations to Easton2 345.0 kV	216.8	429	33.1
Park Ave to Yanke 138.0 kV	Vienna Operations to Easton 2 Substation 345.0 kV	207.0	381	27.2
Yanke Substation to Jarratt 138.0 kV Line	Easton 2 to Vienna Operations 345.0 kV	205.0	388	27.2
Huntington Falls to Hubbell 138.0 kV	Southtown to Commonwealth Chesapeake 345.0 kV	196.7	186	12.4
Arcgen to Chincoteague 138.0 kV line	Commonwealth Chesapeake to Southton 345.0 kV	170.6	197	16.1
Henry St Wps to Quinn 138.0 kV	Vienna Operations to Easton 2 345.0 kV	170.4	425	25.0
Cannelton Hydroelectric Plant to Henry St Wps 138.0 kV	Vienna Operations to Easton 2 345.0 kV	165.8	449	25.0
Menomonie to Wattsville 138.0 kV	Southtown to Commonwealth Chesapeake 345.0 kV	164.2	302	24.0
Cannelton Hydroelectric Plant to Equus Freeport Power 138.0 kV	Vienna Operations Substation to Easton 2 345.0 kV	160.0	485	25.0
Menomonie Substation to Commonwealth Chesapeake 138.0 kV	Southtown Substation to Commonwealth Chesapeake 345.0 kV	159.6	321	24.0

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Equus Freeport Power to Knight St 138.0 kV	Vienna Operations to Easton 2 345.0 kV Line	157.0	506	25.0
Chincoteague to Wattsville 138.0 kV line	Commonwealth Chesapeake to Southtown 345.0 kV	156.6	326	24.0
Persimmon Gap Substation to Sollers Warf 138.0 kV	Vienna Operations to Easton 2 345.0 kV	155.9	547	32.3
Knight Street to Long Lake 138.0 kV line	Vienna Operations to Easton 2 345.0 kV	152.3	542	25.0
Long Lake to Mcminnville 138.0 kV L	Easton 2 to Vienna Operations 345.0 kV	151.2	551	25.0
Saddlebrooke Ranch to Lmmers 138.0 kV	Vienna Operations to Easton 2 345.0 kV Line	150.7	752	47.7
Linde to Mcminnville 138.0 kV line	Easton 2 to Vienna Operations 345.0 kV L	150.2	560	25.0
Linde to Ripple Substation 138.0 kV line	Vienna Operations to Easton 2 345.0 kV	149.2	569	25.0
Ess Lewes to Rehoboth 138.0 kV	Vienna Operations to Easton 2 345.0 kV	146.1	385	25.0
Cady to Dodd Road 345.0 kV	Easton 2 to Navy Mole 138.0 kV	141.6	1,246	64.7
Rodessa to Hubbell 138.0 kV	Commonwealth Chesapeake to Southtown 345.0 kV	140.5	215	12.4
Memphis Jct to Lammers 138.0 kV	Vienna Operations to Easton 2 Substation 345.0 kV line	140.3	637	32.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Vienna Operations 138/1kV TX	Replace transformer with higher-rated unit	\$4,684,000	30% (\$1,389,000)
Southtown to Vienna Operations 345.0 kV	Rebuild line with higher-capacity conductor	\$69,341,000	27% (\$18,920,000)
Prunedale to Valleyview Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$12,484,000	35% (\$4,384,000)
Henning to Kiwi 138.0 kV	Rebuild line with higher-capacity conductor	\$5,926,000	28% (\$1,650,000)
Park Ave to Yanke 138.0 kV	Rebuild line with higher-capacity conductor	\$8,443,000	26% (\$2,190,000)
Yanke Substation to Jarratt 138.0 kV Line	Rebuild line with higher-capacity conductor	\$19,124,000	26% (\$4,961,000)
Huntington Falls to Hubbell 138.0 kV	Rebuild line with higher-capacity conductor	\$7,155,000	31% (\$2,206,000)
Arcgen to Chincoteague 138.0 kV line	Rebuild line with higher-capacity conductor	\$8,604,000	29% (\$2,483,000)
Henry St Wps to Quinn 138.0 kV L	Rebuild line with higher-capacity conductor	\$1,915,000	26% (\$497,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cannelton Hydroelectric Plant to Henry St Wps 138.0 kV	Rebuild line with higher-capacity conductor	\$4,009,000	26% (\$1,040,000)
Menomonie to Wattsville Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$9,222,000	31% (\$2,846,000)
Equus Freeport Power to Cannelton Hydroelectric Plant 138.0 kV	Rebuild line with higher-capacity conductor	\$3,569,000	26% (\$926,000)
Menomonie Substation to Commonwealth Chesapeake 138.0 kV	Rebuild line with higher-capacity conductor	\$9,226,000	31% (\$2,847,000)
Equus Freeport Power to Knight St 138.0 kV	Rebuild line with higher-capacity conductor	\$2,927,000	26% (\$759,000)
Wattsville to Chincoteague 138.0 kV Line	Rebuild line with higher-capacity conductor	\$6,965,000	31% (\$2,148,000)
Persimmon Gap Substation to Sollers Warf 138.0 kV	Rebuild line with higher-capacity conductor	\$20,063,000	24% (\$4,863,000)
Knight St to Long Lake 138.0 kV line	Rebuild line with higher-capacity conductor	\$13,777,000	26% (\$3,573,000)
Long Lake to McMinnville 138.0 kV	Rebuild line with higher-capacity conductor	\$15,417,000	26% (\$3,998,000)
Lammers to Saddlebrooke Ranch 138.0 kV	Rebuild line with higher-capacity conductor	\$10,223,000	25% (\$2,580,000)
Linde to McMinnville 138.0 kV L	Rebuild line with higher-capacity conductor	\$7,289,000	26% (\$1,891,000)
Linde to Ripple 138.0 kV	Rebuild line with higher-capacity conductor	\$9,542,000	26% (\$2,475,000)
Rehoboth to Ess Lewes 138.0 kV line	Rebuild line with higher-capacity conductor	\$6,617,000	26% (\$1,716,000)
Cady to Dodd Road 345.0 kV	Rebuild line with higher-capacity conductor	\$55,105,000	23% (\$12,480,000)
Rodessa to Hubbell 138.0 kV	Rebuild line with higher-capacity conductor	\$6,090,000	33% (\$2,028,000)
Memphis Jct to Lammers 138.0 kV	Rebuild line with higher-capacity conductor	\$15,376,000	24% (\$3,727,000)
	Total allocated to project		\$88,577,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.